RAMAIAH
Institute of Technology

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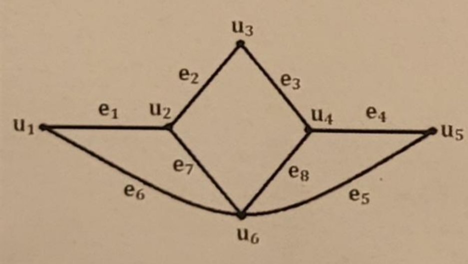
DEPARTMENT OF MATHEMATICS

Sub Code:	AI 34/AD 34/CI 35/CY 35	**Sub:**	Discrete mathematical Structures	**Test:**	II
Time:	3.00 pm-4.00 pm.	**Term:**	20.11.23 to 2.03.24	**Marks:**	30
Date:	27.02.2024	**Semester:**	III	**Sections:**	AI/AD/CI/CY

Note: Answer any TWO full questions. Each main question carries 15 marks.

Q.No.		**Questions**	**CO's**	**Blooms level**	**Marks**
1.	(a)	Solve the recurrence relation $a_n = 7a_{n-1} - 12a_{n-2}$, $a_0 = 3, a_1 = 11$.	CO2	L1	5
	(b)	Draw the graph of the following adjacency matrix and hence write its incidence matrix. $A = \begin{array}{c@{}c} & \begin{array}{ccccc} u & v & w & x & y \end{array} \\ \begin{array}{c} u \\ v \\ w \\ x \\ y \end{array} & \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix} \end{array}$	CO3	L3	5
	(c)	Prove that a tree with *n* vertices has *n*− 1 edges.	CO4	L3	5
2.	(a)	Consider $D_{70} = \{1,2,5,7,10,14,35,70\}$, the divisors of 70 with divisibility as order relation. Verify whether it is a Boolean algebra or not?	CO2	L2	5
	(b)	Find (i) Fusion of u_1 and u_2 (ii) $G - e_1$ (iii) $G - u_1$ for the following graph. 	CO3	L4	5

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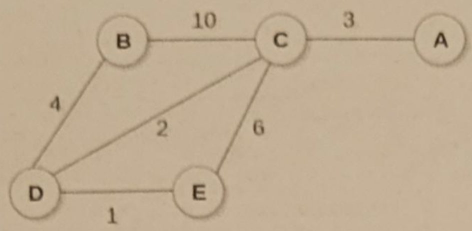
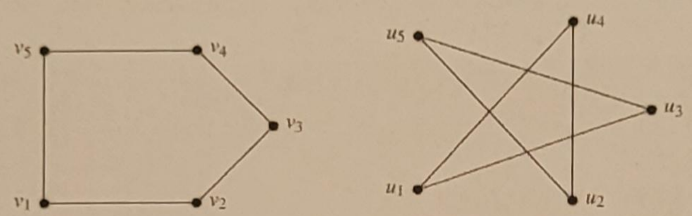
	(c)	Write Prim's algorithm to find the shortest spanning tree and hence find the minimal spanning tree of the following graph: 	CO4	L3	5
3.	(a)	Find an explicit formula for the sequence $a_n = a_{n-1} + 3^n$ given $a_1 = 4$.	CO2	L4	5
	(b)	Define (i) walk (ii) path in a graph G. (iii) Verify whether the following graphs are isomorphic or not?  (a) G_1. (b) G_2.	CO3	L2	5
	(c)	Define semi group. Let G be the set of real numbers excluding 1 and * be defined by $a*b = a+b-ab$ for $a,b \in G$ Check whether G is monoid or not?	CO5	L3	5